

App test (40³, sliding, 20 GPa, 1 MPa)

E_app
u field
 σ field
element von Mises
element |u|
surface p99
Pistoia factor
force (equilibrium)

Tapia et al. protocol (40³, fixed, 18 GPa, 100 N)

E_app
u field
 σ field
element von Mises
element |u|
surface p99
Pistoia factor
force (equilibrium)

Periodic tensor (32³)

tensor C (max.)
log-Euclidean distance

Convergence (22-40³)

E_app (series)

Progressive failure (32³)

E_app per step
failure F per step

10⁻¹⁴ 10⁻¹² 10⁻¹⁰ 10⁻⁸ 10⁻⁶ 10⁻⁴

|relative difference| spinpy vs FEBio

VOI Spinodoid Dual-lattice declared tolerance